PROPERTY F IMPLIES WEAK INTEGRALITY FOR PSEUDO-UNITARY BRAIDED FUSION CATEGORIES

Abstract. We give a proof by contradiction that every braided fusion category with property F and a pseudo-unitary Galois conjugate is weakly integral. The proof combines a trace inequality for finite braid group images with Galois symmetry in the Drinfeld centre. A non-pseudo-unitary conjugate would produce a Galois image of the unit whose conjugated dimension lies strictly between zero and one. The trace inequality forces the absolute value of this dimension to equal the object's Frobenius–Perron dimension, which is at least one.

1. Introduction

Let $\mathcal{C}$ be a braided fusion category over $\mathbb{C}$. For an object X and $n \geq 2$, write $G_{X,n}$ for the image of

$$\rho_{X,n} : B_n \longrightarrow \mathrm{Aut}(X^{\otimes n}), \qquad \sigma_i \longmapsto \mathrm{id}_X^{\otimes(i-1)} \otimes c_{X,X} \otimes \mathrm{id}_X^{\otimes(n-i-1)}.$$

The category has *property F* if every $G_{X,n}$ is finite, and is *weakly integral* if $\mathrm{FPdim}(\mathcal{C}) \in \mathbb{Z}$.

Conjecture (Naidu–Rowell [NR])**.** *A braided fusion category has property F if and only if it is weakly integral.*

Thus the conjecture relates finiteness of all braid group images to a single arithmetic condition on the fusion category. We prove the implication from property F to weak integrality under a pseudo-unitarity hypothesis. Recall that $\mathcal{C}$ is *pseudo-unitary* if $\dim(\mathcal{C}) = \mathrm{FPdim}(\mathcal{C})$.

Theorem 1.1. *Let $\mathcal{C}$ be a braided fusion category with a pseudo-unitary Galois conjugate. If $\mathcal{C}$ has property F, then $\mathcal{C}$ is weakly integral.*

The proof takes place in the Drinfeld centre of a pseudo-unitary Galois conjugate of $\mathcal{C}$. We show that every further Galois conjugate remains pseudo-unitary; its global dimension is then fixed by Galois conjugation and hence is an integer. If a conjugate were not pseudo-unitary, Galois symmetry in the centre would produce a simple object Y whose dimension in a conjugate category lies strictly between zero and one. Finite braid images force the absolute value of this dimension to equal $\mathrm{FPdim}(Y)$, which is at least one.

The main step is the trace argument in Section 3. It first bounds the categorical trace of every unitary in a braid algebra. A particular three-strand braid attains this bound; the equality case then gives the required identity of dimensions. Section 4 places the relevant Galois images of the unit in a subcategory of the centre with property F and completes the contradiction. It then derives a criterion for a single object to have infinite braid images.

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2. Preliminaries

All fusion categories are over $\mathbb{C}$, and we suppress associativity and unit constraints. We write $\mathbf{1}$ for the tensor unit and $N_Z^W = \dim \mathrm{Hom}(W, Z)$ for the multiplicity of a simple object W in an object Z.

2.1. **Transfer of property F.** Property F passes to direct summands and fusion subcategories. Indeed, if V is a summand of X, naturality of the braiding makes $V^{\otimes n}$ invariant under the braid action on $X^{\otimes n}$, and restriction gives a quotient of $G_{X,n}$. It also passes along a dominant braided tensor functor: the braid image on the image of an object is a quotient of its original braid image, and every object of the target is a summand of such an image.

If $\mathcal{C}$ has property F, so do $\mathcal{C}^{\mathrm{rev}}$ and $\mathcal{C} \boxtimes \mathcal{C}^{\mathrm{rev}}$. Reversing the braiding inverts the braid generators. On $X \boxtimes Y$, the braid image is a quotient of a subgroup of $G_{X,n} \times G_{Y,n}$, and every object of the product category is a summand of some $X \boxtimes Y$.

Finally, property F and Frobenius–Perron dimensions are invariant under Galois conjugation. A number-field model for the braided categorical data exists by [ENO, 2.30, 2.33]. Conjugation preserves all relations among braid matrices and preserves the fusion rules. We regard Galois automorphisms as automorphisms of $\overline{\mathbb{Q}}$ over $\mathbb{Q}$.

2.2. **Dimensions.** In a spherical category, d_X denotes categorical dimension, Tr categorical trace, and tr ordinary matrix trace. Simple dimensions and their Galois conjugates are real and nonzero [ENO, 2.3, 2.5, 2.10]. The dimension function is a character of the fusion ring, as is each of its Galois conjugates. Thus $\sigma(d_X)$ is a fusion-matrix eigenvalue, and Perron–Frobenius theory gives

$$|\sigma(d_X)| \le \mathrm{FPdim}(X).$$

Since $\dim(\mathcal{C}) = \sum_X d_X^2$, global dimensions are totally positive and $\dim(\mathcal{C}) \le \mathrm{FPdim}(\mathcal{C})$ [ENO, 8.21]. On a pseudo-unitary category we use the canonical spherical structure, for which $d_X = \mathrm{FPdim}(X)$ [ENO, 8.23]. On a Galois conjugate we retain the transported spherical structure.

Remark (Spherical structures under conjugation). We record why the chosen spherical structure is defined over a finite extension of the field of definition, so that it can be transported under Galois conjugation. For each simple X, choose a nonzero algebraic map $a_X : X \to X^{**}$ and write its pivotal component as $j_X = t_X a_X$. Let $q_X(a_X)$ denote the quantum trace defined using the duality maps. This trace is algebraic and nonzero [ENO, Section 2.1], and

$$d_X = t_X q_X(a_X).$$

Since d_X is algebraic, so is t_X. There are only finitely many simple objects, giving the required finite extension.

2.3. **Galois symmetry of modular data.** A modular category is a non-degenerate braided fusion category with a spherical structure. Let $\mathcal{M}$ be modular, put $D = \dim(\mathcal{M})$, and set

$$S_{XY} = \mathrm{Tr}_{X \otimes Y}(c_{Y,X} c_{X,Y}).$$

The matrix S is symmetric and $S^2 = DE$, where $E_{XY} = \delta_{X,Y^*}$. Its normalized columns $\chi_Y(X) = S_{XY}/d_Y$ are fusion-ring characters. Non-degeneracy makes them distinct, so they exhaust the characters [EGNO, 8.13.11, 8.14.2]. Thus every Galois

automorphism τ determines a permutation $\widehat{\tau}$ of the simple labels by $\tau \circ \chi_Y = \chi_{\widehat{\tau}(Y)}$. This permutation is distinct from passage to the conjugate category, whose tensor unit is still $\mathbf{1}$.

Proposition 2.1. *If $Y = \widehat{\tau}(\mathbf{1})$, then Y is self-dual and*

$$S_{XY} = d_Y \tau(d_X) \quad \text{for every simple } X, \tag{1}$$

$$\tau(D) = D/d_Y^2. \tag{2}$$

Proof. The first identity is $\chi_Y = \tau \circ \chi_{\mathbf{1}}$. Therefore

$$d_Y^2 \tau(D) = \sum_X S_{XY}^2 = (S^2)_{YY} = D\delta_{Y,Y^*}.$$

The left side is nonzero, proving both assertions. □

3. The trace argument

Fix an object X for which every braid image $G_{X,n}$ is finite, and let $A_n \subseteq \mathrm{End}(X^{\otimes n})$ be the linear span of $G_{X,n}$. Averaging makes the group action unitary on each multiplicity space. The resulting adjoint makes A_n a finite-dimensional $*$-algebra, with

$$\left(\sum_{g \in G_{X,n}} \alpha_g g \right)^* = \sum_{g \in G_{X,n}} \overline{\alpha_g} g^{-1}.$$

Thus the involution is independent of the averaged inner products. Tensoring on either side with identity strands preserves it, since it carries braid images and their inverses to braid images and inverses.

The next lemma obtains a trace bound by placing overlapping copies of a unitary on increasingly many strands.

Lemma 3.1 (Trace bound)**.** *Let X be simple in a spherical braided fusion category, with $G_{X,n}$ finite for every n. Put $d = d_X$ and $f = \mathrm{FPdim}(X)$. Every unitary $u \in A_m$, $m \geq 2$, satisfies*

$$|\operatorname{Tr}(u)| \leq |d| f^{m-1}.$$

Proof. Place k copies $u_1, \ldots, u_k$ on consecutive blocks of m strands, successive blocks sharing one strand, so $N = k(m-1)+1$. These insertions preserve the involution, so $U = u_1 \cdots u_k$ is unitary. On each multiplicity space its ordinary trace has absolute value at most the dimension of that space. Consequently,

$$\begin{aligned} |\operatorname{Tr}(U)| &= \left| \sum_W d_W \operatorname{tr}\Big(U|_{\mathrm{Hom}(W, X^{\otimes N})}\Big) \right| \\ &\leq \sum_W N_{X^{\otimes N}}^W \, |d_W| \leq \sum_W N_{X^{\otimes N}}^W \, \mathrm{FPdim}(W) = f^N. \end{aligned}$$

Simplicity of X makes the partial trace of u over its last $m-1$ strands equal to $(\operatorname{Tr}(u)/d)\,\mathrm{id}_X$. Only u_k meets the last $m-1$ strands of U; tracing them out and iterating gives

$$\operatorname{Tr}(U) = \frac{\operatorname{Tr}(u)^k}{d^{k-1}}, \qquad |\operatorname{Tr}(u)|^k \leq |d|^{k-1} f^{k(m-1)+1}.$$

Take k-th roots and let $k \to \infty$. □

For the rest of this section, let $\mathcal{M}$ be pseudo-unitary and modular, let τ be a Galois automorphism, and put

$$Y = \widehat{\tau}(\mathbf{1}), \qquad \sigma = \tau^{-1}, \qquad f = d_Y = \mathrm{FPdim}(Y), \qquad d = \sigma(d_Y).$$

We now work in $\mathcal{M}^\sigma$, retaining the simple-object labels of $\mathcal{M}$. The dimensions there are $\sigma(d_W)$, where d_W denotes the original dimension in $\mathcal{M}$. All traces, braid operators, and multiplicity spaces below are taken in $\mathcal{M}^\sigma$, and A_n denotes the span of $G_{Y,n}$. These braid images are isomorphic to those in $\mathcal{M}$ by Galois invariance.

Lemma 3.2 (Galois rigidity)**.** *If $G_{Y,n}$ is finite for every n, then $f = |d|$.*

Proof. We first bound the total absolute weight of the categorical trace on A_3. Decompose the multiplicity spaces into irreducible representations of the finite group $G_{Y,3}$:

$$H_W = \mathrm{Hom}(W, Y^{\otimes 3}) \cong \bigoplus_r V_r^{\oplus m_{Wr}}, \qquad A_3 \cong \bigoplus_r \mathrm{End}(V_r),$$

retaining only the constituents that occur. For $T \in A_3$,

$$\mathrm{Tr}(T) = \sum_r w_r \,\mathrm{tr}(T|_{V_r}), \qquad w_r = \sum_W m_{Wr}\sigma(d_W).$$

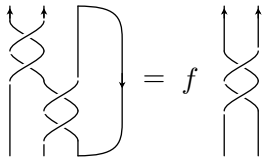
The multiplicities m_{Wr} are nonnegative integers, and for each retained r at least one is positive. Since the original dimensions d_W are positive,

$$w_r = \sigma\left(\sum_W m_{Wr} d_W\right) \neq 0.$$

The weights are real. Let $J \in A_3$ act on V_r by $\mathrm{sgn}(w_r)$. This is a unitary involution, so Lemma 3.1 gives

$$\sum_r |w_r| \dim V_r = \mathrm{Tr}(J) \leq |d| f^2. \tag{3}$$

We now exhibit a braid whose trace attains this bound. Let a, b be the images of σ_1, σ_2, and put $c = c_{Y,Y}$. In $\mathcal{M}$, equation (1) gives $S_{YY} = f\tau(f)$; applying σ yields $\mathrm{Tr}(c^2) = df$ in $\mathcal{M}^\sigma$. Since Y is simple, the partial trace of c^2 over its second strand is $f\,\mathrm{id}_Y$. Tracing over the third strand of a^2b^2 therefore gives



Partial trace over strand 3.

$$\mathrm{Tr}(a^2b^2) = f\,\mathrm{Tr}(c^2) = df^2.$$

Since $g = a^2b^2$ is unitary,

$$|d|f^2 = \left|\sum_r w_r \,\mathrm{tr}(g|_{V_r})\right| \leq \sum_r |w_r|\,|\,\mathrm{tr}(g|_{V_r})| \leq \sum_r |w_r| \dim V_r \leq |d| f^2.$$

Equality holds throughout. Since every w_r is nonzero, each unitary matrix $g|_{V_r}$ has trace of absolute value $\dim V_r$, and hence is scalar. Hence a^2b^2 is central in A_3. Commuting it with b shows that a^2 commutes with b; it also commutes with a, so it is central. The braid generators are conjugate, whence $a^2 = b^2$. Thus

$$c^2 \otimes \mathrm{id}_Y = \mathrm{id}_Y \otimes c^2.$$

Taking the partial trace over the third strand yields $d\,c^2 = f\,\mathrm{id}_{Y\otimes Y}$. Since c^2 has finite order, $|f/d| = 1$. □

Remark. Pseudo-unitarity is used here to identify d_Y with $\mathrm{FPdim}(Y)$ and to ensure that every trace weight w_r is nonzero. The conjugated dimension $d = \sigma(d_Y)$ may be negative, which explains the absolute value in the conclusion.

4. The main theorem and consequences

To apply Lemma 3.2, we place the Galois images of the unit in a modular category where their braid images are finite. The required category is the Drinfeld centre; the relevant objects lie in a subcategory generated by the canonical images of $\mathcal{C}$ and $\mathcal{C}^{\mathrm{rev}}$.

For a fusion subcategory $\mathcal{A}$ of a braided category, write $\mathcal{A}'$ for its centralizer. Let

$$\iota_+ : \mathcal{C} \longrightarrow Z(\mathcal{C}), \quad X \longmapsto (X, c_{X,-}), \qquad \iota_- : \mathcal{C}^{\mathrm{rev}} \longrightarrow Z(\mathcal{C}), \quad X \longmapsto (X, c^{-1}_{-,X}),$$

and denote their images by $\mathcal{C}_+$ and $\mathcal{C}_-$.

Lemma 4.1. *Let $\mathcal{C}$ be a pseudo-unitary braided fusion category with property F. Then $\mathcal{M} = Z(\mathcal{C})$ is pseudo-unitary and modular, and $\mathcal{B} = \mathcal{C}_+ \vee \mathcal{C}_-$ has property F and contains $\widehat{\tau}(\mathbf{1})$ for every Galois automorphism τ.*

Proof. The centre is modular. Its global and Frobenius–Perron dimensions are $\dim(\mathcal{C})^2$ and $\mathrm{FPdim}(\mathcal{C})^2$, respectively, so it is pseudo-unitary [EGNO, 7.16.6, 8.20.14, 9.3.4]. The canonical images are mutual centralizers inside $\mathcal{M}$: (V, γ) centralizes $\mathcal{C}_+$ precisely when $\gamma_X = c^{-1}_{X,V}$ for every X, and similarly for $\mathcal{C}_-$. Consequently,

$$\mathcal{B}' = \mathcal{C}_+ \cap \mathcal{C}_- = \iota_+(\mathcal{C}').$$

This category is symmetric, hence integral [DGNO, 2.46]. For $Y = \widehat{\tau}(\mathbf{1})$ and every simple $X \in \mathcal{B}'$, we therefore have $d_X = \mathrm{FPdim}(X) \in \mathbb{Z}$, and

$$S_{XY} = d_Y \tau(d_X) = d_Y d_X.$$

The S-matrix criterion for centralizers and the double-centralizer theorem give $Y \in \mathcal{B}'' = \mathcal{B}$ [EGNO, 8.20.5, 8.21.1]. Finally, the two canonical images centralize each other, so $X \boxtimes Y \mapsto \iota_+(X) \otimes \iota_-(Y)$ defines a braided functor $\mathcal{C} \boxtimes \mathcal{C}^{\mathrm{rev}} \to \mathcal{B}$ [EGNO, 8.6.1]. It is dominant because these images generate $\mathcal{B}$. The transfer properties in Section 2.1 therefore give property F for $\mathcal{B}$. □

Proof of Theorem 1.1. By Galois invariance we may assume that $\mathcal{C}$ is pseudo-unitary. Put $\delta = \dim(\mathcal{C}) = \mathrm{FPdim}(\mathcal{C})$. We show that every Galois conjugate of $\mathcal{C}$ is pseudo-unitary. Their common fusion rules will then force every Galois automorphism to fix δ, proving integrality. Suppose, to the contrary, that $\mathcal{C}^\sigma$ is not pseudo-unitary. Since Galois conjugation preserves fusion rules,

$$0 < \sigma(\delta) = \dim(\mathcal{C}^\sigma) < \mathrm{FPdim}(\mathcal{C}^\sigma) = \delta.$$

Set $\mathcal{M} = Z(\mathcal{C})$, $D = \delta^2$, $\tau = \sigma^{-1}$ and $Y = \widehat{\tau}(\mathbf{1})$. By Lemma 4.1, all braid images of Y are finite. Write $f = d_Y = \mathrm{FPdim}(Y)$ and $d = \sigma(d_Y)$. Equation (2) and the positivity of δ and $\tau(\delta) = \dim(\mathcal{C}^\tau)$ give

$$f = \sqrt{\frac{D}{\tau(D)}} = \frac{\delta}{\tau(\delta)}, \qquad d = \frac{\sigma(\delta)}{\delta} \in (0, 1).$$

Lemma 3.2 now gives the contradiction

$$1 \leq \mathrm{FPdim}(Y) = f = |d| = d < 1.$$

Thus every Galois conjugate of $\mathcal{C}$ is pseudo-unitary, and every Galois automorphism fixes δ. As δ is an algebraic integer, $\delta \in \mathbb{Z}$. □

The theorem also has an objectwise consequence. To apply it to the fusion subcategory $\langle X \rangle$ generated by an object X, we must establish property F for every object of that subcategory. The following lemma shows that finiteness of all $G_{X,n}$ suffices.

Lemma 4.2 (Closure)**.** *Let X be an object of a braided fusion category. If $G_{X,n}$ is finite for every n, then $\langle X \rangle$ has property F.*

Proof. The case $X = 0$ is immediate. Otherwise, summands of finite sums of positive powers of X form a nonzero full abelian subcategory closed under tensor products and subquotients. By [EGNO, 4.11.4], it contains the unit and is rigid, so it equals $\langle X \rangle$. Thus every object of $\langle X \rangle$ is a summand of some $Z = \bigoplus_i X^{\otimes k_i}$, with $k_i \geq 1$. For each $n \geq 2$, the pure braid group P_n preserves each summand $X^{\otimes k_{i_1}} \otimes \cdots \otimes X^{\otimes k_{i_n}}$ of $Z^{\otimes n}$. By cabling, its action there has image in the finite group $G_{X,k_{i_1}+\cdots+k_{i_n}}$. Thus its image on $Z^{\otimes n}$ is finite, being contained in a finite product of finite groups. Since $[B_n : P_n] = n!$, the full braid image is finite as well [ERW, 2.1(ii)]. Naturality passes finiteness to every summand of Z, as in Section 2.1. □

Consequently, the dimension of a particular simple object can force its own braid images to be infinite.

Corollary 4.3. *Let X be a simple object of a braided fusion category, and suppose that $\langle X \rangle$ has a pseudo-unitary Galois conjugate. If $\mathrm{FPdim}(X)^2 \notin \mathbb{Z}$, then $G_{X,n}$ is infinite for every sufficiently large n.*

Proof. If every $G_{X,n}$ were finite, Lemma 4.2 and Theorem 1.1 would make $\langle X \rangle$ weakly integral. Then $\mathrm{FPdim}(X)^2 \in \mathbb{Z}$ by [ENO, 8.27], a contradiction. Hence G_{X,n_0} is infinite for some n_0. Tensoring with a nonzero object is faithful, so tensoring with identity strands embeds G_{X,n_0} in $G_{X,n}$ for every $n \geq n_0$. □

In particular, the conclusion holds in every unitary braided fusion category, since its fusion subcategories are unitary and hence pseudo-unitary. The simplicity hypothesis is essential: direct sums in a weakly integral category need not have integral squared Frobenius–Perron dimension.

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